Bioinformatics, 2026, pp. 1–7

doi: DOI HERE
Advance Access Publication Date: Day Month Year
Original Paper

ORIGINAL PAPER

TrackPilot: choosing which genomic foundation-model tracks to trust for a cell type from a few measured variants

[TODO: First Author][1,*] and [TODO: Co-author][1]

[1][TODO: Department], [TODO: Institution], [TODO: Country]
*Corresponding author. [TODO: email]
FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

Abstract

Motivation: Sequence-to-function models such as AlphaGenome predict the effect of a variant on about a thousand cell-type-specific tracks. A laboratory studying one cell type must decide which of these predictions to use. Matching tracks by cell-type name is unreliable, and many cell types are not profiled at all. Laboratories often measure a few variants before a larger study, but there is no principled way to choose those variants or to turn a handful of measurements into a choice among a thousand correlated tracks.
Results: We present TrackPilot. It treats each track as a hypothesis that the assay is an unknown linear function of the track's predictions, integrates the calibration out in closed form, and starts from the tracks whose metadata match the cell type. It proposes a pilot panel of variants on which the plausible tracks disagree, and returns a posterior-weighted prediction for every unmeasured variant. We evaluated it in pre-registered tests on three mutagenesis reporter assays. On 125 loci, eight pilot measurements were non-inferior to the matched tracks (mean +0.010 in Spearman correlation) and, with these tracks hidden, improved on the all-track mean by 0.047 ($P = 3 \times 10^{-12}$); eight random measurements made the same inference worse than not measuring. These results held on a second dataset, with checkpoints not trained on the tested loci and with a second model, Enformer. In cell types without tracks of their own the gains were not consistent: pilot panels still beat random ones, but eight measurements did not beat the all-track mean in mouse cell lines, and in induced neurons no track predicted the assay. A permutation check on the same measurements then falls back to the label-free default and recovers part of the loss.
Availability and implementation: TrackPilot is a Python package and command-line tool (MIT licence) at `https://github.com/ydvislab/AuraGenome-V` [TODO: final repository, PyPI name, Zenodo DOI of the submitted version]. Pre-registrations and all derived results are in the same repository.
Contact: [TODO: email]
Supplementary information: Supplementary data are available at *Bioinformatics* online.

Key words: variant effect prediction, genomic foundation models, active model selection, Bayesian experimental design, massively parallel reporter assays

1. Introduction

Deep sequence-to-function models predict the regulatory consequences of non-coding variants from DNA sequence alone [Zhou and Troyanskaya, 2015, Avsec et al., 2021, Linder et al., 2025]. AlphaGenome, the most recent of them, predicts thousands of genome tracks for hundreds of human cell types and tissues and can score any single-nucleotide variant on all of them [Avsec et al., 2026]. On saturation mutagenesis benchmarks such as the CAGI5 regulatory challenge [Kircher et al., 2019, Shigaki et al., 2019] it is among the best available predictors. Its output, however, is not one prediction per variant but about a thousand, one per track.

A laboratory that studies variants in a particular cell type therefore has to choose which tracks to use. The usual choice is

© The Author 2026. Published by Oxford University Press. All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

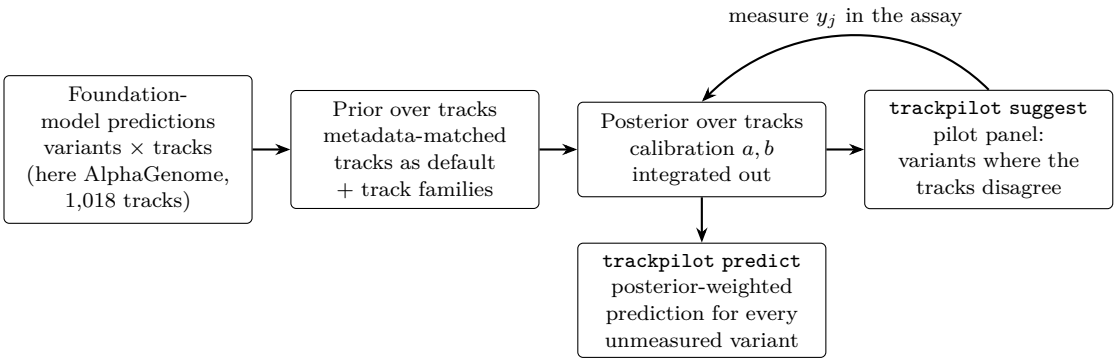


Fig. 1. TrackPilot. Each track is a hypothesis that the measured effect is an affine function of the track's predicted effect. Starting from a prior that favours the metadata-matched tracks, TrackPilot proposes a panel of variants on which the tracks disagree, updates the posterior with their measured effects and predicts every other variant. [TODO: redraw as a two-panel figure: (a) the problem, tracks disagreeing in sign at one locus; (b) the workflow.]

the tracks whose biosample name matches the cell line. We found this unreliable. Across the 1,018 CAGE, DNase and ATAC tracks of AlphaGenome, the name-matched track ranked between first and 27th by agreement with measured effects on our development elements, and the choice changed biological conclusions: at the *PKLR* promoter in K562 cells, replacing K562 tracks by HepG2 tracks lowered the fraction of significant variants with the correct sign from 0.96 to 0.69–0.85. Many cell types used in the laboratory, including patient-derived lines, are not profiled, and then there is no name to match. Adapting the model to a new cell type is possible but requires genome-wide profiles of that cell type [Hingerl et al., 2025].

What a laboratory usually can do is measure a few variants, for example in a pilot reporter assay of 8–16 constructs before a larger study. This turns the choice of tracks into a label-efficient model selection problem: decide which variants to measure so that the measurements identify the tracks, or the mixture of tracks, that best predict the rest [Cohn et al., 1996]. Active model selection methods address this for classifiers [Karimi et al., 2021, Okanovic et al., 2025, Kay et al., 2025]. Two features of foundation-model tracks break their assumptions. The tracks are regression outputs on arbitrary scales, so each must be calibrated to the assay with an unknown slope and offset that a few labels cannot pin down. And they are a thousand strongly correlated heads of one network, so a few labels often cannot settle which one is best [Kuboyama, 2026], and averaging over the plausible tracks accounts for this uncertainty [Hoeting et al., 1999]. The problem is not specific to AlphaGenome: any model with many output heads and a user with a few labels faces it.

Here we present TrackPilot, a tool that turns a few measured variants into a decision about which tracks to use. TrackPilot (i) treats each track as a hypothesis with an affine calibration that is integrated out in closed form; (ii) starts from the metadata-matched tracks and moves away from them only when the measurements contradict them; (iii) chooses the variants to measure by the classical criterion for discriminating between models [Box and Hill, 1967]; and (iv) reports a posterior-weighted prediction rather than one selected track. We evaluated it in pre-registered tests on three mutagenesis reporter datasets and asked three questions in turn: is it worse than the matched tracks when they exist, does it find a substitute when they are missing, and where does it stop helping. We also compared it with simpler panels, few-label regression and active model selection, and checked the dependence on the model's training exposure.

2. Methods

2.1. Problem

A regulatory element is studied in cell type c. A foundation model gives a predicted effect $x_{t,j}$ for each candidate variant j on each of T tracks (T = 1,018 here). The laboratory can measure the effects y_j of k variants, chosen from a pool of variants that can be built, together with the unmutated reference sequence as a control, so that each effect is measured relative to the reference. We want a prediction $\hat{y}_j$ for every other variant, and the panel S, $|S| = k$, that makes these predictions as accurate as possible. We measure accuracy by the Spearman correlation between $\hat{y}$ and y on variants outside the pool.

2.2. Tracks as calibrated hypotheses

Each track's predictions are scaled over the element but not centred, $z_{t,j} = x_{t,j}/\mathrm{sd}_j(x_{t,\cdot})$, so that the sign keeps the direction of the alternative allele. Hypothesis t states

$$\begin{aligned} y_j &= a + b\, z_{t,j} + \varepsilon_j, \\ a &\sim N(0, \alpha s^2), \quad b \sim N(0, \beta s^2), \quad \varepsilon_j \sim N(0, (1-\beta)s^2), \end{aligned} \tag{1}$$

where s^2, the mean of y_S^2, is a plug-in estimate of the label scale. Given it, integrating out a and b gives the marginal likelihood of the measured panel,

$$y_S \mid t \sim N\Big(0,\ s^2\left[(1-\beta)I + \alpha \mathbf{1}\mathbf{1}^\top + \beta z_{t,S} z_{t,S}^\top\right]\Big), \tag{2}$$

and the posterior $p(t \mid y_S) \propto p(t)\, p(y_S \mid t)$. The predictive mean $m_{t,j} = E[y_j \mid t, y_S]$ follows from the Gaussian conditional, and TrackPilot reports the Bayesian model average $\hat{y}_j = \sum_t p(t \mid y_S)\, m_{t,j}$ [Hoeting et al., 1999]. The two prior parameters were fixed before any evaluation ($\alpha = 0.1$, $\beta = 0.5$) and never tuned.

2.3. Metadata prior

When the model profiles the cell type, the mean of the metadata-matched tracks is a strong default. TrackPilot adds it as one extra hypothesis that holds prior mass $\pi = 0.95$, with a slope prior centred on a positive value, $b \sim N(\sqrt{\beta}\, s,\ (\beta/4)\, s^2)$, so that a negative slope has prior probability of about 2%. Single tracks share the remaining mass. Tracks whose predictions correlate at $r \geq 0.9$ form a family (average linkage, no labels used), and the prior is split equally between families and then within them, so a family of near-duplicates shares the prior mass of one track. Without matched

tracks there is no default and all mass goes to single tracks. π was chosen on the Kircher development elements and frozen before the Siraj test; unlike the pre-registered tests, the rule for this choice was not committed before the development runs.

2.4. Choosing the pilot panel

The panel is built sequentially from the pool. Given the current posterior, the next variant maximises the posterior-weighted variance of the hypotheses' predictive means, $\mathrm{Var}_{t\sim p(t|y_S)}[m_{t,j}]$, divided by the expected residual variance. This is the Box–Hill criterion for discriminating between models [Box and Hill, 1967, Rainforth et al., 2024]. Before any label, all predictive means are zero under the symmetric slope prior, so the first step uses each track's standardised prediction. Because the choice depends only weakly on the measured values (Section 3.3), TrackPilot can also return the whole panel before any measurement, by replacing each chosen variant's label with the current posterior-average prediction.

2.5. Number of measurements

The panel size k is set by the user (default 8). We fixed $k = 8$ before the tests as a practical budget, the smallest usual pilot size (8–16 constructs), not as a derived minimum, and report k from 2 to 16. With very few labels each track's calibration is poorly determined, so the posterior can move on noise rather than on differences between tracks; the learning curves (Section 3.3) show where this matters. Further rounds can be added.

2.6. Applicability check

TrackPilot can only help if some track, or mixture of tracks, follows the assay. After the panel is measured, TrackPilot compares the mixture's log evidence, $\log \sum_t p(t)\, p(y_S \mid t)$, with its value for 200 random permutations of the same labels. If the labels are explained no better than shuffled ones (permutation $P > 0.5$, a threshold chosen on development units), TrackPilot returns the label-free default instead. This is an empirical fallback rule, not a test of absent signal; with a panel chosen sequentially from the labels, as in our evaluation, its null is approximate.

2.7. Implementation

TrackPilot is a Python package with a command-line interface [TODO: finalise once the package is released; keep in step with the README]. Its input is a table of predicted effects (variants $\times$ tracks) with track metadata, so it is not tied to one model; an optional module scores a region with AlphaGenome using the user's own API key (not yet run from the package with a live key). `trackpilot suggest` returns a pilot panel of k variants from a pool; Its size is set with `-k`, and `--measured` adds a further round to labels already measured. `trackpilot predict` takes any number of measured effects and returns the posterior over tracks, predictions for all variants, the prediction after the applicability check and a report. A 1,353-variant element with 1,018 tracks takes under five seconds on a laptop. A reproduction script recomputes with the package every frozen TrackPilot result for the first repeat (22 Kircher experiments, 234 numbers; 125 Siraj loci in two cell types, 4,500 numbers) with no difference; the pre-registered Kircher version (uniform prior) is not covered.

2.8. Data and predictions

Kircher et al. [Kircher et al., 2019] measured every possible substitution in 20 disease-associated regulatory elements by saturation mutagenesis MPRA. We used the 22 experiments in human cells (27,789 variants). Five experiments from five regions formed the development set; the other 17 experiments from 11 regions formed the test set. The six experiments in mouse cell lines (Neuro-2a, MIN6, NIH/3T3; five regions) were analysed only in a separate pre-registered test. **Siraj et al.** [Siraj et al., 2026, 2025] measured all substitutions in 200-bp oligonucleotides around fine-mapped trait variants in K562 and HepG2 cells. We used 125 of the 128 published loci (one excluded for a deletion in the reference, two for having no unique match in GRCh38), 74,975 substitutions each measured in both cell types. The effect is the $\log_2$ fold change of the substituted oligo minus that of the unmutated reference oligo in the same experiment, which a laboratory measuring a pilot panel would also build. **Kosicki et al.** [Kosicki et al., 2025] measured synthetic transversions at every fourth base of 270-bp tiles in iPSC-derived excitatory neurons (ENCSR548AQS): 328 tiles with at least 30 substitutions (22,397 measurements) in 159 regions, effects relative to the reference tile. Four AlphaGenome and two Enformer tracks match neurons. **Predictions.** We scored every variant by in-silico mutagenesis with AlphaGenome (1-Mb context, default distilled checkpoint) on all CAGE, DNase and ATAC tracks for human biosamples [Avsec et al., 2026], which derive from ENCODE and FANTOM5 [ENCODE Project Consortium et al., 2020, The FANTOM Consortium and the RIKEN PMI and CLST (DGT), 2014]. We scored the neuron regions in the same way, and for Section 3.5 the 11 Kircher test regions, the mouse-cell regions, 40 randomly drawn Siraj loci and the neuron regions with Enformer [Avsec et al., 2021] on its 1,322 human CAGE, DNase and ATAC tracks (196-kb context, one forward pass), using the same score: the change in $\log_2$ of the summed signal over a 501-bp window centred on the variant.

2.9. Evaluation design

Scenarios. We hid tracks to simulate three situations: *profiled*, all tracks kept; *related-only*, the element's own cell line hidden; *unprofiled*, the own and related cell types hidden. Related cell types were listed from track metadata before any result was seen. These scenarios simulate a missing cell type; cell types that truly lack tracks are analysed separately (Section 3.7). **Protocol.** In each of 200 repeats, half of an element's variants form the pool that could be measured. Every method is scored by Spearman correlation on the same variants outside the pool. The unit of analysis is the region (Kircher; experiments in one region averaged) or the locus (Siraj; two cell types averaged). **Pre-registration.** Each test below was committed with frozen code before its test data were analysed. For Kircher (version without the metadata prior): H1, the pilot panel beats random panels; H2, it beats the all-track mean by at least 0.03 with a positive difference in at least 7/11 regions. For Siraj (frozen TrackPilot): P1, non-inferiority to the matched mean in the profiled scenario at $k = 8$ (one-sided 95% bootstrap lower bound above -0.02); P2, at most 15% of loci worse by more than 0.05; P3, improvement over the all-track mean when unprofiled at $k = 8$ of at least 0.03; P4, improvement over the related-cell mean when related-only at $k = 16$. For the neurons (N1–N4) and the mouse-cell experiments (M1–M3), the analogues of these hypotheses at $k = 8$ (Supplementary Tables S10, S14); for the applicability check, A1 and A2 (Section 3.8). Analyses added later are labelled post hoc. **Training data, test sets and homology.** TrackPilot trains no model; AlphaGenome is used frozen, and the only labels are the k measured variants of the element being predicted. No measurement from another element and no evaluation variant enters the inference. The test sets were fixed before analysis and do not overlap the development regions. AlphaGenome's training tracks do cover these regions, so we repeated the tests with, for each locus,

Table 1. Pre-registered Siraj test, frozen TrackPilot, 125 loci (two cell types averaged per locus). D is the mean paired difference in Spearman correlation. S rows are secondary analyses.

Hyp.	Comparison	D	Criterion, result
P1	profiled, $k = 8$: TrackPilot − matched mean	+0.010	LB $+0.005 > -0.02$, holds
P2	profiled, $k = 8$: loci with $D < -0.05$	3/125	UB $0.061 \leq 0.15$, holds
P3	unprofiled, $k = 8$: TrackPilot − all-track mean	+0.047	≥ 0.03, $P = 3\times10^{-12}$, holds
P4	related-only, $k = 16$: TrackPilot − related mean	+0.040	> 0, $P = 3\times10^{-19}$, holds
S1	unprofiled, $k = 8$: pilot − random panel	+0.096	$122/125 > 0$
S2	profiled, $k = 8$: TrackPilot − `meta_ridge`	+0.007	$97/125 > 0$
S3	profiled, $k = 8$: `matched_cal` (random panel) − matched	−0.082	$1/125 > 0$

the checkpoint whose held-out folds contain every training window overlapping the locus (`alphagenome.data.fold_intervals`). These folds are the Borzoi partitions, which assign genome fragments to eight folds at random while keeping human and mouse orthologous regions together [Linder et al., 2025]. Homologous sequence within the human genome is not explicitly grouped, so the held-out-fold results reduce, rather than remove, training exposure. **Baselines.** Label-free: the all-track mean, the matched-track mean and the related-cell mean. Other panels: random, largest predicted effect, and largest unweighted disagreement. Few-label regression: ridge [Hoerl and Kennard, 1970] and LASSO [Tibshirani, 1996] over all tracks; an affine re-fit of the matched mean (`matched_cal`); and a ridge fit on residuals of the calibrated matched mean (`meta_ridge`). Active model selection: CODA- and MODEL SELECTOR-style selection [Kay et al., 2025, Okanovic et al., 2025] with our calibrated likelihood. The best single track chosen with all labels is reported as a reference.

3. Results

3.1. When matched tracks exist, TrackPilot was non-inferior to them on Siraj and Kircher

In a profiled cell type the matched tracks are the natural default, so the first requirement is not to do worse than them. On Siraj, with all tracks kept, eight measurements were non-inferior to the matched mean (P1: +0.010, one-sided lower bound +0.005 against a margin of −0.02), and only 3/125 loci were worse by more than 0.05 (P2: upper bound 6.1% against 15%; Table 1, Fig. 2). At $k = 16$ the difference was +0.019 with one locus harmed.

The metadata prior is what keeps TrackPilot level with the matched tracks. Without it, TrackPilot was 0.029 below the matched mean on the Kircher test regions at $k = 8$, with 7/11 regions harmed (post hoc): it helped where the matched tracks were poor (*LDLR*: matched mean 0.24, TrackPilot 0.51) but lost where they were good (*SORT1*: 0.65 against 0.48). With the prior, chosen on development elements before these runs, it was level with the matched mean (+0.001, one region harmed; +0.018 and none harmed at $k = 16$). The obvious shortcut, re-fitting the matched mean to the same few labels, lowered accuracy on Siraj (−0.082 on random panels, −0.010 on pilot panels), because the fitted slope is poorly determined and can flip sign. A ridge fit on residuals of the matched mean (`meta_ridge`) was close to TrackPilot (−0.007; Table 1, S2).

3.2. When matched tracks are missing, eight measurements find a substitute

With the element's own and related cell types hidden, the label-free alternative is the mean of all tracks. On Siraj, eight measurements improved on it by 0.047 (0.391 to 0.438; $P = 3 \times 10^{-12}$; P3). With related-cell tracks only, sixteen measurements improved on their mean by 0.040 (0.412 to 0.452; $P = 3 \times 10^{-19}$; P4). The Kircher test, with different elements, cell lines and assay design, agreed: eight pilot measurements raised the mean Spearman correlation from 0.296 (all-track mean) to 0.370, while eight random measurements gave 0.286 (Table 2; Supplementary Fig. S1). Both pre-registered hypotheses held: the pilot panel beat random panels by 0.084 (9/11 regions, $P = 0.0024$) and the all-track mean by 0.074 (8/11 regions).

The gain was concentrated where the default was poor (post hoc): in the quarter of Siraj loci where the all-track mean was furthest from the best track it was 0.133, more than half of that gap, and in the quarter where it was closest, 0.004 (Supplementary Table S8). It was absent in elements with very few significant variants (*FOXE1*, 5). Figure 3 shows one run, on the *IRF4* enhancer in SK-MEL-28 melanoma cells with the melanocyte and related tracks hidden. After eight measurements the posterior favoured DNase tracks of neural crest cells, from which melanocytes derive, and the prediction reached $\rho = 0.59$ on held-out variants, against 0.39 for the all-track mean.

3.3. Which and how many variants are measured matters; one round matched sequential choice

Measuring random variants was worse than not measuring: in the unprofiled scenario on Siraj, the same inference reached 0.342 with random panels, below the label-free all-track mean (0.391), and the pilot panel beat random panels at 122/125 loci (S1). The specific criterion mattered less: choosing by the expected information gain about which track is correct matched the pilot panel (differences ≤ 0.003; post hoc; Supplementary Tables S4, S6). Panels of the largest predicted effects were as good when matched tracks were kept, but worse without them (Siraj unprofiled: −0.028, 97/125 loci). Most useful in practice, a panel chosen in one round, before any measurement, was within 0.005 of sequential choice on both datasets, inside our pre-declared equivalence margin of 0.02. A laboratory can therefore order the whole panel at once. Accuracy rose with k (Siraj, matched tracks hidden, post hoc; Supplementary Table S7): two measurements were below the label-free default (0.347 against 0.391), four were above it (0.402) and as accurate as sixteen random ones (0.404), and eight and sixteen reached 0.438 and 0.461. With matched tracks, a stronger default, eight measurements were needed to pass it (0.483 against 0.474; four: 0.469).

3.4. Comparison with regression and active model selection

With the same eight labels, regression over all tracks was much worse than TrackPilot (Table 2), and on the pilot panel itself TrackPilot's inference still beat LASSO and ridge by 0.023–0.048. We adapted the active model selection methods CODA and MODEL SELECTOR [Kay et al., 2025, Okanovic et al., 2025], gave them our calibrated likelihood, and fixed CODA's prior weight on development elements

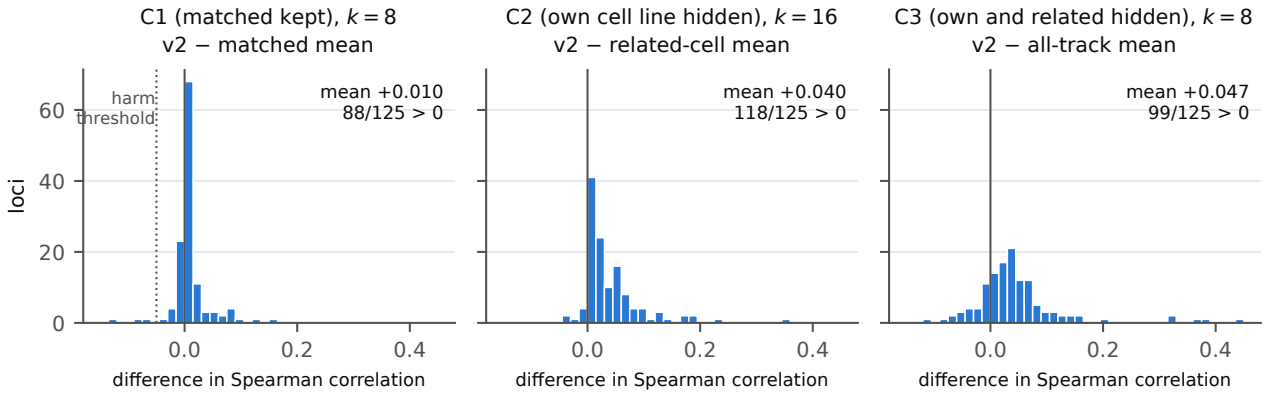


Fig. 2. Per-locus paired differences on the 125 Siraj loci for the pre-registered comparisons. Dotted line: the pre-registered harm threshold (−0.05).

Table 2. Kircher test set, unprofiled scenario (own and related cell types hidden), k = 8, 11 regions. Spearman correlation on held-out variants. Pre-registered version (uniform prior); the metadata prior is inactive in this scenario. †Post hoc (first 50 repeats); for LASSO the range covers three rules for all-zero fits.

Panel	Prediction	Mean ρ	TrackPilot − this row	Regions > 0, one-sided P
pilot (TrackPilot)	track posterior	0.370	—	—
random	track posterior	0.286	+0.084	9/11, 0.0024
largest predicted effect	track posterior	0.338	+0.031	–, 0.18
none	all-track mean	0.296	+0.074	8/11, –
random	LASSO†	–	+0.110 to +0.182	9–10/11, ≤0.0024
random	ridge†	–	+0.116	8/11, 0.0068
pilot	LASSO†	–	+0.034 to +0.048	8/11, ≤0.016
pilot	ridge†	–	+0.023	8/11, 0.042
all labels	best single track (reference)	0.454	—	—

(post hoc; Supplementary Table S5). TrackPilot beat the CODA-style selected track by 0.075 on Siraj with matched tracks kept (111/125 loci), by 0.029–0.034 without them, and by 0.051–0.082 on Kircher. Reporting the baseline's posterior average instead of its single track recovered 0.016–0.026; with matched tracks hidden TrackPilot kept a small advantage over the averaged baseline (Siraj +0.018, 95% interval 0.011–0.024). The advantage thus comes from the metadata prior and from averaging over calibrated tracks.

3.5. A second foundation model

TrackPilot uses only the matrix of predictions and the track metadata, so it should carry over to other models. We re-ran the frozen method, with no parameter changed, on Enformer predictions. The hypotheses and thresholds, the same as in the AlphaGenome tests, were committed before scoring; Siraj was restricted to 40 randomly drawn loci for compute. All six hypotheses held (Supplementary Table S15). With matched tracks, eight measurements were non-inferior to them (+0.022, one-sided lower bound +0.007; 1/40 loci worse by more than 0.05). Without them, eight measurements improved on the all-track mean by 0.052 on Siraj ($P = 4 \times 10^{-4}$) and by 0.109 on Kircher (8/11 regions), and beat eight random measurements by 0.097 (10/11 regions, $P = 0.001$). Enformer's tracks were less accurate than AlphaGenome's, and TrackPilot's gains were at least as large. Both models' training data cover these regions and include K562 and HepG2, so this test addresses a change of model, not of cell type.

3.6. Robustness

Training exposure. The default checkpoint was distilled from teachers trained on all training folds, so it has seen tracks over every tested locus [Avsec et al., 2026]. We repeated both tests with, for each locus, the AlphaGenome checkpoint that held out its genomic fold, using reading rules committed before the runs. Absolute accuracies fell by 0.035–0.07, for the matched tracks most (Siraj: 0.474 to 0.402). All four Siraj hypotheses still held, with larger effects (P1 +0.024, P3 +0.050, P4 +0.053; Supplementary Table S3). On the 11 Kircher regions the pilot panel's advantage over random panels grew (+0.093, 10/11, $P = 0.0015$) and its advantage over the all-track mean shrank (+0.054, $P = 0.062$). **Label definition.** The published centring of the Siraj effects changed each contrast by ≤ 0.002 (Supplementary Table S1). **Direction of effects.** On Siraj sign accuracy agreed with the rank results; on Kircher TrackPilot called fewer signs correctly than the matched mean (−0.133; −0.071 without the fitted offsets, which it therefore also reports; Supplementary Table S2). **Sharing across loci.** In a pre-registered extension, a prior pooled from ten other loci of the same cell type added 0.012 at eight measurements on 85 test loci with matched tracks hidden, below the pre-registered 0.02, and stayed non-inferior with them (Supplementary Table S9).

3.7. Cell types without tracks of their own

Hiding the tracks of a profiled cell type usually leaves a closely related track. Three tests address cell types with no tracks of their own.

Five Kircher test regions were measured in human cell lines without tracks of their own (HEL 92.1.7, SK-MEL-28, three

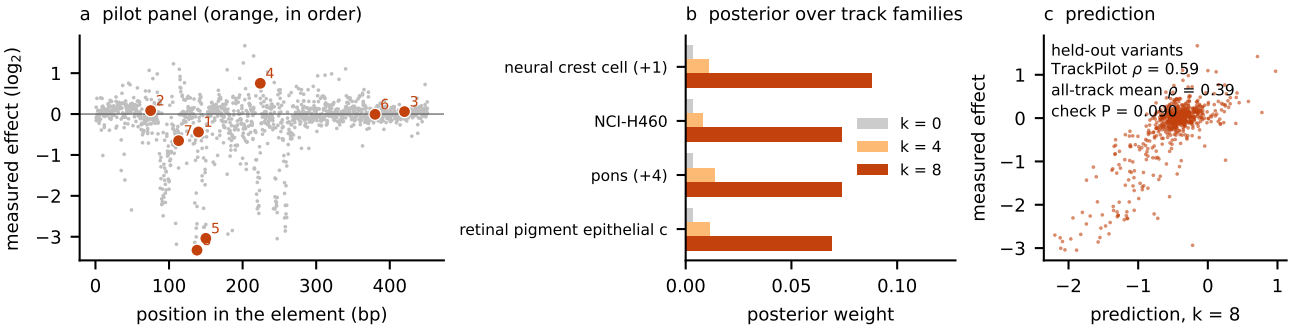


Fig. 3. Worked example: *IRF4* enhancer in SK-MEL-28, melanocyte and related tracks hidden, AlphaGenome, one repeat. (a) Measured effects; pilot panel in orange, in order. (b) Posterior weight of the four leading track families after 0, 4 and 8 measurements. (c) Held-out variants against the prediction.

glioblastoma lines), with a related biosample, named before any run, as their proxy. There (post hoc), TrackPilot kept the proxy's accuracy (+0.002 with AlphaGenome, +0.010 with Enformer) and, with the proxy hidden, beat the all-track mean by 0.12 and 0.16 (3/5 and 4/5 regions) and random panels by 0.10–0.11 (5/5; Supplementary Table S13).

We then pre-registered a test on the five Kircher regions measured in mouse cell lines (Neuro-2a, MIN6, NIH/3T3), which no human track profiles, with human proxies named from metadata. It was not supported with either model (Supplementary Table S14). With the proxy hidden, pilot panels beat random ones (AlphaGenome +0.086, Enformer +0.102) but not the all-track mean (−0.004, +0.022; required 0.03). With the proxy, TrackPilot corrected a poor one (*ZFAND3*: +0.17, +0.14) but lost 0.068 at *RET* with AlphaGenome, where the proxy was already close to the best track. The best single track reached 0.30–0.34, so a signal was there, but eight measurements did not find it reliably; with sixteen (post hoc) the difference to the all-track mean grew to +0.019 and +0.040.

In induced neurons, which the models barely profile, the pre-registered test (N1–N4, both models) failed for a different reason: no track predicted the assay. The matched tracks reached a Spearman correlation of 0.07 and the all-track mean 0.04–0.05; TrackPilot was 0.02–0.03 below the matched mean and 0.01–0.04 below the all-track mean, with about 30% of regions worse by more than 0.05 (Supplementary Table S10).

3.8. A fallback when the measurements follow no track

A user cannot see the tracks' accuracy before measuring, but can compare the fit of the eight labels with that of shuffled copies. We therefore pre-registered the applicability check, evaluated with sequentially chosen panels: its threshold was chosen on development units of all three datasets (5 Kircher experiments, 40 Siraj loci, 50 neuron regions) and committed before the held-out units were run. On the 109 held-out neuron regions it fell back in 43–46% of runs, raised accuracy over unchecked TrackPilot by 0.010 (65/109 regions, $P = 0.002$; A1) and stayed within 0.004 of the default. On the 85 held-out Siraj loci it intervened in 5–8% of runs and kept the gains: +0.056 over the all-track mean without matched tracks and +0.011 over the matched mean with them (A2; Supplementary Table S11). The overall neuron results had been seen before the check was designed, so A1 is weaker evidence. With Enformer predictions (secondary) the check gained 0.019 over unchecked TrackPilot but stayed 0.015 below the default: here the permutation P values were close to uniform, so about half of the runs passed by chance. A stricter threshold (0.2, an option of TrackPilot), equally good on development units, cut this loss to at most 0.007 for both models and cost at most 0.004 on Siraj (post hoc; Supplementary Table S12). On the *IRF4* run of Fig. 3 the check passed ($P = 0.09$).

4. Discussion

TrackPilot gives laboratories a concrete protocol for using a foundation model in their own cell type: start from the tracks whose metadata match, measure about eight variants chosen to split the competing tracks, and let the measurements move the prediction only when they contradict the default. The evidence for this protocol has three tiers. When matched tracks exist, it was non-inferior to them on average in rank correlation on Siraj and Kircher, though not in neurons, where no track predicted the assay. When they are missing but tracks of other cell types carry the signal, it recovered 0.04–0.05 in Spearman correlation, with two datasets, two models and held-out checkpoints. In cell types without tracks of their own it still chose better variants than random, but eight measurements did not reliably beat the all-track mean, and where no track predicted the assay the applicability check fell back to the default in about half the runs. The number of measurements is the user's main lever: with two to four labels the prediction can fall below the default, and beyond eight accuracy kept rising, most where no close track exists, which is where a second round pays.

Three lessons appear general: which variants are measured matters as much as how they are modelled; re-fitting a default predictor to a handful of labels is harmful, and on Kircher and Siraj a prior centred on it avoided these losses; and among a thousand correlated heads, a posterior average beats the single most probable track.

Limitations. The main gains were measured by hiding the tracks of profiled cell types. For cell types without tracks of their own the only positive evidence is post hoc and covers five regions; the pre-registered test in mouse cell lines was not supported. With eight labels the applicability check has limited power. The evaluation is retrospective, but a panel chosen in one round, before any label, was as accurate (Section 3.3), so a laboratory can order the panel at once; the fallback rule was evaluated only with sequential panels. Non-inferiority on average does not exclude losses at single loci (3/125 worse by more than 0.05, the largest −0.13; −0.26 in one cell type). Absolute correlations are moderate (0.44–0.49 on Siraj). TrackPilot helps only if at least one track, or a mixture of tracks, is roughly proportional to the assay.

5. Conclusion

A handful of well-chosen measurements can guide which of a thousand foundation-model tracks to rely on for a cell type, as long as the decision starts from the metadata default, the measurements are chosen to discriminate, and some track close to the cell type carries the assay's signal. For cell types far from every profiled one, more measurements or new profiles are needed. TrackPilot makes this a two-command step: one to choose the panel and one to turn its measurements into predictions, with a fallback to the default when the measurements fit no track better than shuffled labels.

Acknowledgements

[TODO: acknowledgements] [TODO: AI-use declaration (journal LLM policy); the authors must rewrite the text.]

Author contributions

[TODO: CRediT roles for each author]

Funding

[TODO: funding]

Conflict of interest

None declared.

Data availability

Kircher et al. data are available from the original publication [Kircher et al., 2019] [TODO: database accession]. Siraj et al. saturation mutagenesis data are available from Zenodo under CC-BY-NC [Siraj et al., 2025]; they are not redistributed. The neuron MPRA data are ENCODE experiment ENCSR548AQS (files ENCFF958ZLG and ENCFF752NFF, CC BY 4.0) [Kosicki et al., 2025]. Predictions can be regenerated with the scripts in the repository. Pre-registrations, code and all derived tables are at `https://github.com/ydvislab/AuraGenome-V`.

References

Žiga Avsec, Vikram Agarwal, Daniel Visentin, et al. Effective gene expression prediction from sequence by integrating long-range interactions. *Nature Methods*, 18:1196–1203, 2021. doi: 10.1038/s41592-021-01252-x.

Žiga Avsec, Natasha Latysheva, Jun Cheng, Guido Novati, et al. Advancing regulatory variant effect prediction with AlphaGenome. *Nature*, 649:1206–1218, 2026. doi: 10.1038/s41586-025-10014-0.

G. E. P. Box and W. J. Hill. Discrimination among mechanistic models. *Technometrics*, 9:57–71, 1967. doi: 10.1080/00401706.1967.10490441.

David A. Cohn, Zoubin Ghahramani, and Michael I. Jordan. Active learning with statistical models. *Journal of Artificial Intelligence Research*, 4:129–145, 1996. doi: 10.1613/jair.295.

ENCODE Project Consortium, Federico Abascal, Reyes Acosta, et al. Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature*, 583:699–710, 2020. doi: 10.1038/s41586-020-2493-4.

Johannes C. Hingerl, Laura D. Martens, Alexander Karollus, Trevor Manz, Jason D. Buenrostro, Fabian J. Theis, and Julien Gagneur. scooby: modeling multimodal genomic profiles from DNA sequence at single-cell resolution. *Nature Methods*, 22(11): 2275–2285, 2025. doi: 10.1038/s41592-025-02854-5.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1): 55–67, 1970. doi: 10.1080/00401706.1970.10488634.

Jennifer A. Hoeting, David Madigan, Adrian E. Raftery, and Chris T. Volinsky. Bayesian model averaging: a tutorial. *Statistical Science*, 14:382–417, 1999. doi: 10.1214/ss/1009212519.

Mohammad Reza Karimi, Nezihe Merve Gürel, Bojan Karlaš, Johannes Rausch, Ce Zhang, and Andreas Krause. Online active model selection for pre-trained classifiers. In *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021. arXiv:2010.09818.

Justin Kay, Grant Van Horn, Subhransu Maji, Daniel Sheldon, and Sara Beery. Consensus-driven active model selection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. arXiv:2507.23771.

Martin Kircher, Chenling Xiong, Beth Martin, et al. Saturation mutagenesis of twenty disease-associated regulatory elements at single base-pair resolution. *Nature Communications*, 10:3583, 2019. doi: 10.1038/s41467-019-11526-w.

Michael Kosicki, Dianne Laboy Cintrón, Pia Keukeleire, et al. Massively parallel reporter assays and mouse transgenic assays provide correlated and complementary information about neuronal enhancer activity. *Nature Communications*, 16:4786, 2025. doi: 10.1038/s41467-025-60064-1.

Tetsuji Kuboyama. How many labels does model choice need? Certificates and budgets for selective prediction. arXiv:2609.18622, 2026.

Johannes Linder, Divyanshi Srivastava, Han Yuan, Vikram Agarwal, and David R. Kelley. Predicting RNA-seq coverage from DNA sequence as a unifying model of gene regulation. *Nature Genetics*, 57:949–961, 2025. doi: 10.1038/s41588-024-02053-6.

Patrik Okanovic, Andreas Kirsch, Jannes Kasper, Torsten Hoefler, Andreas Krause, and Nezihe Merve Gürel. All models are wrong, some are useful: Model selection with limited labels. In *Proceedings of the 28th International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025. arXiv:2410.13609.

Tom Rainforth, Adam Foster, Desi R. Ivanova, and Freddie Bickford Smith. Modern Bayesian experimental design. *Statistical Science*, 39:100–114, 2024. doi: 10.1214/23-STS915.

Dustin Shigaki, Orit Adato, Aashish N. Adhikari, Shengcheng Dong, et al. Integration of multiple epigenomic marks improves prediction of variant impact in saturation mutagenesis reporter assay. *Human Mutation*, 40:1280–1291, 2019. doi: 10.1002/humu.23797.

Layla Siraj, Rodrigo Ivan Castro Lopez Vaal, Hannah Dewey, Susan Kales, et al. Functional dissection of complex and molecular trait variants at single nucleotide resolution [dataset]. Zenodo, 2025.

Layla Siraj, Rodrigo I. Castro, Hannah B. Dewey, Susan Kales, et al. Functional dissection of complex trait variants at single-nucleotide resolution. *Nature*, 2026. doi: 10.1038/s41586-026-10121-6.

The FANTOM Consortium and the RIKEN PMI and CLST (DGT). A promoter-level mammalian expression atlas. *Nature*, 507:462–470, 2014. doi: 10.1038/nature13182.

Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, 58:267–288, 1996. doi: 10.1111/j.2517-6161.1996.tb02080.x.

Jian Zhou and Olga G. Troyanskaya. Predicting effects of noncoding variants with deep learning-based sequence model. *Nature Methods*, 12:931–934, 2015. doi: 10.1038/nmeth.3547.